

ENGR 4411: Physical Acoustics

Homework Assignment 4

Due: Monday, April 20 at midnight.

Problems marked with [*] are potential sources for quiz questions.

Problem 1. Rüchardt's Method¹: A simple harmonic oscillator that combines a gas spring and a (nearly) frictionless piston (Airpot[®]) is shown in Figure 1. The glass cylinder that contains the piston is sealed to a platform that contains a microphone that can measure and record the oscillating pressure within the cylinder if the piston is displaced from its equilibrium position and allowed to execute simple harmonic motion with frequency, $\omega_0 = 2\pi f_0 = \sqrt{K_{gas}/m}$.

The spring constant, K_{gas} , of a “gas spring” comprised of a cylinder of uniform cross-sectional area, A , and volume, $V = Ah$, depends upon the mean pressure, p_m , of the gas contained within the cylinder.

$$K_{gas} = \frac{\gamma p_m}{V} A^2 = \rho_m c^2 \frac{A^2}{V},$$

A , $V = Ah$, and p_m can be measured accurately. The value of ω_0 can be determined from the period of oscillation, T .

The text author used a ruler and a marker pen to place marks separated by 1.0 cm along the glass cylinder. The microphone and the PVC protector excluded some of the volume within the cylinder. I assumed the compressible volume was determined by the measured height of the piston plus some “fudge” length, h_0 , that would take care of any excluded volumes and any offset in my marking of the cylinder's length. The value of the polytropic coefficient of air, γ_{air} , could then be determined by a least-squares fit of the square of the measured periods, T_i^2 , plotted against the equilibrium heights of the piston, h_i , based on the marks on the cylinder

$$\left[\frac{p_m A}{4\pi^2 m} \right] T_i^2 = \frac{1}{\gamma_{air}} (h_i + h_0)$$

The polytropic coefficient would therefore be equal to the reciprocal of the slope, $\gamma_{air} = (\text{slope})^{-1}$, and the “height offset” h_0 , would be the ratio of the intercept to the slope, $h_0 = (\text{intercept})/(\text{slope})$. The data in Table 1 was acquired at an atmospheric pressure of 97.3 kPa. (Is that the mean pressure, p_m , within the volume?) Using your favorite method and plotting tools determine γ_{air} and the relative uncertainty in slope of the best-fit line. The accepted value for the polytropic coefficient of dry air is $\gamma_{air} = 1.403$. What is the relative difference between your experimental value of γ_{air} and the accepted value? If that relative difference is larger than the relative uncertainty in the slope, suggest at least one source of systematic error that could be responsible for that discrepancy.

Problem 2. Speed of Sound: You sometimes see the speed of sound in air written as,

$$c(\text{m/s}) = 331 + 0.6T(^{\circ}\text{C}).$$

Show how this can be derived by starting with $c = \sqrt{\gamma r T}$ near the freezing point of water, $T = 273\text{K}$. Hint: expand the expression for c in a Taylor Series for T .

Problem 3. Addition of three incoherent sound sources: At a particular shop, three machines produce individual sound pressure levels of 90, 93, and 95 dB_{SPL}, all referenced to 20 $\mu\text{Pa}_{\text{rms}}$. Determine the total

¹E. Rüchardt, Eine einfache methode zur bestimmung von C_p/C_v . Phys. Z. **30**, 58-59 (1929).

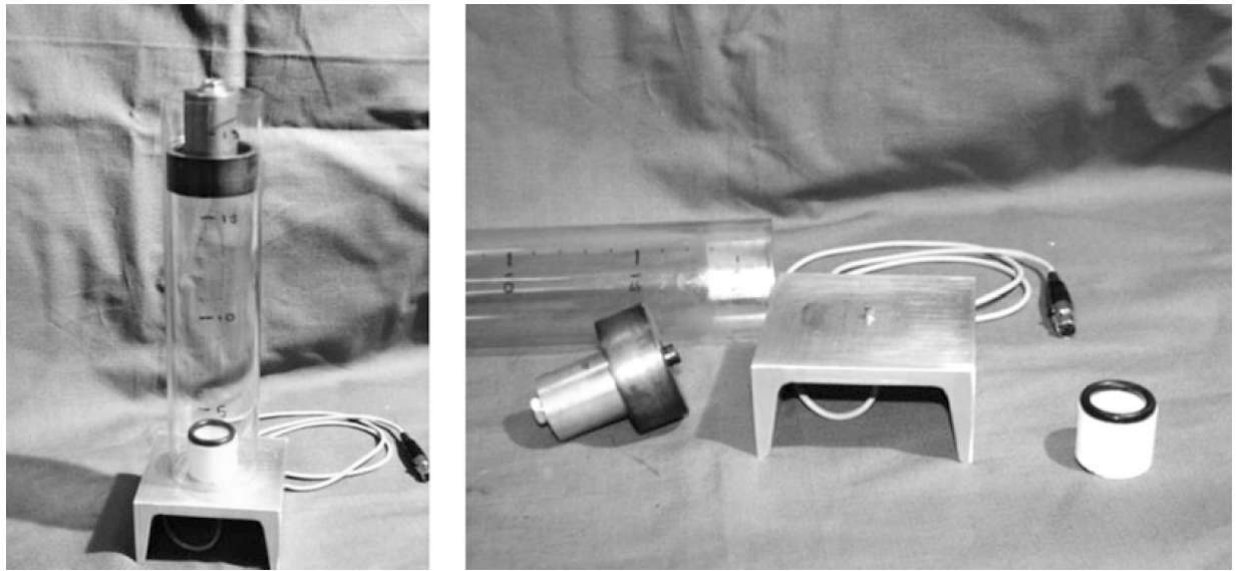


Figure 1: (Left) This simple harmonic oscillator is made from an AirpotTM Precision Air Dashpot, consisting of a glass cylinder and a tight-fitting graphite piston of diameter 44.33 ± 0.03 mm that acts as a gas spring with stiffness, K_{gas} . The graphite piston and brass cylinder provide a total piston mass, $m = 184.41$ grams. The bottom end of the glass cylinder is sealed to an aluminum platform with vacuum grease. (Right) An Endevco piezoresistive pressure sensor is threaded into the platform and is protruding through the platform. The microphone's electrical output is connected to a digital storage oscilloscope to allow accurate measurement of the natural period of the freely decaying oscillations. The small (white) PVC tube with the rubber O-ring on top is placed over the microphone as protection against any possible collision between the microphone and the graphite piston.

sound pressure level if all of the machines are running simultaneously. Assume each noise source is statistically independent of all the others so that their powers (not pressures) are combined. Report your result in dB re: $20 \mu\text{Pa}_{\text{rms}}$.

[*] **Problem 4. Climate change:** Table 10.1 shows standard dry air having a CO_2 mole fraction of 314 ppm. As of June 2020, the mole fraction was found to have increased to 415 ppm.² How much does this change the speed of sound in dry air?

- (a) *Approximate sound speed change.* Get an approximate result by ignoring the change in γ . ie only consider the change in the molecular mass, M_{mix} , of dry air.
- (b) *Polytropic coefficient.* Use $\gamma_{\text{air}} = 1.4$ and $\gamma_{\text{CO}_2} = 1.3$. Will the change in the sound speed due to the change in γ_{mix} have the same sign as that due to the change in M_{mix} ?
- (c) *Sound speed change.* Use Eqn 10.25 to calculate the change in γ , and use that result to obtain a more accurate result for the change in sound speed for dry air.

[*] **Problem 5. Schlagwetter-pfeife:** The presence of methane in mines presents a significant hazard due to the possibility of underground explosions. In the late-1880s, one of the world's most famous industrial chemists, Fritz Haber,³ invented the "methane whistle" to determine the presence of hydrogen or methane in air extracted from underground mines. By 1888, these methane whistles also began to appear in the United Kingdom. Air

²The 2026 value for the mole fraction of CO_2 is 429.4 ± 0.6 ppm.

³Fritz Haber was also one of the world's most infamous chemists, having developed the gas that was used to murder prisoners in the Nazi death camps.

Heights(h_i) (m)	Periods (T_i) (milliseconds)
0.15	70.480
0.13	65.400
0.19	79.725
0.18	77.600
0.15	70.567
0.14	68.143
0.13	65.680
0.11	60.467
0.10	57.880
0.08	50.450
0.19	79.800
0.18	77.867
0.17	75.267
0.16	73.200
0.15	70.850
0.14	68.467

Table 1: oscillation periods T_i , in milliseconds, for various equilibrium heights, h_i , of the bottom of the graphite piston measured using the apparatus as shown in Figure 1.

was pumped from the mine out through one whistle, and fresh air was pumped through the other. Assume the whistles were identical and in good thermal contact, so both surface and subsurface air passing through the whistles were at the same temperature. Determine the concentration (mole fraction) of methane (CH_4) in the mine air if the air whistle had a frequency of 440 Hz and the pair of whistle produced 10 beats per second at 20°C .